

Lesson 6-1: Division Expressions with Fractions and Whole Numbers

Connect & apply • 8 tasks • Prepares for Standard 6.NS.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute $8.4 \div 4$.

Answer: 2.1

84 tenths divided by 4 is 21 tenths.

Prerequisite skill: Divide decimals and whole numbers

TASK 2 • FOUNDATION 2

What is the reciprocal of 5?

Answer: $\frac{1}{5}$

Write 5 as $\frac{5}{1}$, then exchange numerator and denominator.

Prerequisite skill: Name the reciprocal of a whole number and of a unit fraction

TASK 3 • FOUNDATION 3

Compute $12 \times \frac{2}{3}$.

Answer: 8

Divide 12 by 3, then multiply by 2.

Prerequisite skill: Multiply a whole number by a fraction

TASK 4 • FOUNDATION 4

Write $\frac{15}{3}$ as a division expression and evaluate it.

Answer: $15 \div 3 = 5$

A fraction bar means divide the numerator by the denominator.

Prerequisite skill: Read a fraction bar as "divided by"

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TASK 5 • QUICK CHECK A

$4.8 \div 2$? How many fourths are in 3? Reciprocal of 5?

Answer: $4.8 \div 2 = 2.4$. There are 12 fourths in 3 (since $3 \div 1/4 = 3 \times 4 = 12$). Reciprocal of 5 is $1/5$.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

$6.4 \div 2$? How many thirds are in 4? Reciprocal of 8?

Answer: $6.4 \div 2 = 3.2$. There are 12 thirds in 4 ($4 \div 1/3 = 4 \times 3 = 12$). Reciprocal of 8 is $1/8$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

$9.6 \div 4$? How many halves are in 5? Reciprocal of $1/6$?

Answer: $9.6 \div 4 = 2.4$. There are 10 halves in 5. Reciprocal of $1/6$ is 6.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A ribbon is 6 feet long. How many $3/4$ -foot pieces can be cut from it?

Answer: $6 \div 3/4 = 6 \times 4/3 = 24/3 = 8$ pieces.

Check the result against the situation and explain your reasoning.

Lesson 6-2: Division Expressions with Fractions and Mixed Numbers

Connect & apply • 8 tasks • Prepares for Standard 6.NS.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Write $2\frac{3}{4}$ as an improper fraction.

Answer: $\frac{11}{4}$

Two wholes contain 8 fourths; add 3 fourths to get 11 fourths.

Prerequisite skill: Rewrite a mixed number as an improper fraction

TASK 2 • FOUNDATION 2

Compute $\frac{2}{3} \times \frac{3}{5}$.

Answer: $\frac{2}{5}$

Multiply to get $\frac{6}{15}$, then simplify by dividing both parts by 3.

Prerequisite skill: Multiply two fractions

TASK 3 • FOUNDATION 3

Find the reciprocal of $\frac{4}{7}$.

Answer: $\frac{7}{4}$

Exchange numerator and denominator; their product will be 1.

Prerequisite skill: Find the reciprocal of any fraction

TASK 4 • FOUNDATION 4

Simplify $\frac{18}{24}$.

Answer: $\frac{3}{4}$

Divide numerator and denominator by their greatest common factor, 6.

Prerequisite skill: Simplify a fraction to lowest terms

Lesson 6-2: Division Expressions with Fractions and Mixed Numbers

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Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

2 $\frac{3}{4}$ as an improper fraction? Reciprocal of $\frac{3}{5}$?
 $\frac{2}{3} \times \frac{5}{4}$?

Answer: 2 $\frac{3}{4}$ as improper fraction: $(2 \times 4 + 3) / 4 = 11/4$. Reciprocal of $\frac{3}{5}$ is $\frac{5}{3}$. Product: $\frac{2}{3} \times \frac{5}{4} = 10/12 = 5/6$.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

3 $\frac{2}{5}$ as an improper fraction? Reciprocal of $\frac{4}{7}$?
 $\frac{3}{4} \times \frac{2}{5}$?

Answer: 3 $\frac{2}{5} = (3 \times 5 + 2)/5 = 17/5$. Reciprocal of $\frac{4}{7}$ is $\frac{7}{4}$. $\frac{3}{4} \times \frac{2}{5} = 6/20 = 3/10$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

1 $\frac{5}{6}$ as an improper fraction? Reciprocal of $\frac{9}{2}$?
 $\frac{5}{6} \times \frac{3}{10}$?

Answer: 1 $\frac{5}{6} = 11/6$. Reciprocal of $\frac{9}{2}$ is $\frac{2}{9}$. $\frac{5}{6} \times \frac{3}{10} = 15/60 = 1/4$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

How many 1 $\frac{1}{2}$ -cup servings are in 6 cups?

Answer: $6 \div \frac{3}{2} = 6 \times \frac{2}{3} = 4$ servings.

Check the result against the situation and explain your reasoning.

Lesson 6-3: Explore Numerical Expressions with Exponents

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.1

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute $3 \times 3 \times 3 \times 3$.

Answer: 81

Successive products are 9, 27, and 81.

Prerequisite skill: Multiply the same factor repeatedly

TASK 2 • FOUNDATION 2

Evaluate 5^2 .

Answer: 25

5^2 means 5×5 .

Prerequisite skill: Recall 2^2 , 3^2 , 4^2 , 5^2 , 2^3 , and 3^3

TASK 3 • FOUNDATION 3

Evaluate $10 \times 10 \times 10$.

Answer: 1000

Three factors of 10 give one thousand.

Prerequisite skill: Multiply by 10 repeatedly to build powers of 10

TASK 4 • FOUNDATION 4

How many factors of 2 appear in $2 \times 2 \times 2 \times 2 \times 2$?

Answer: 5

Count each repeated factor, not the multiplication signs.

Prerequisite skill: Count how many factors are written

Lesson 6-3: Explore Numerical Expressions with Exponents

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Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

6^2 ? Write $2 \times 2 \times 2 \times 2$ as a power. 10^3 ?

Answer: $6^2 = 6 \times 6 = 36$. $2 \times 2 \times 2 \times 2$ as a power = 2^4 . $10^3 = 10 \times 10 \times 10 = 1,000$.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

7^2 ? Write $3 \times 3 \times 3 \times 3 \times 3$ as a power. 10^4 ?

Answer: $7^2 = 49$. $3 \times 3 \times 3 \times 3 \times 3 = 3^5$. $10^4 = 10,000$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

4^3 ? Write $5 \times 5 \times 5$ as a power. 10^2 ?

Answer: $4^3 = 64$. $5 \times 5 \times 5 = 5^3$. $10^2 = 100$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Which is greater, 3^4 or 4^3 ?

Answer: $3^4 = 81$ and $4^3 = 64$, so 3^4 is greater.

Check the result against the situation and explain your reasoning.

Lesson 6-4: Write and Evaluate Numerical Expressions with Exponents

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.2.c

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Evaluate 4^3 .

Answer: 64

$4 \times 4 \times 4 = 16 \times 4 = 64$.

Prerequisite skill: Evaluate a power

TASK 2 • FOUNDATION 2

Evaluate $2 + 3^2 \times 4$.

Answer: 38

Exponent first gives 9; multiplication gives 36; then add 2.

Prerequisite skill: Follow order of operations with parentheses and exponents

TASK 3 • FOUNDATION 3

Evaluate $(8 + 4)/3$.

Answer: 4

The fraction bar groups the numerator: $12 \div 3 = 4$.

Prerequisite skill: Read a fraction bar as a grouping symbol

TASK 4 • FOUNDATION 4

Write an expression for 7 less than the product of 3 and 5.

Answer: $3 \times 5 - 7$

Start with the product, then subtract 7. Its value is 8.

Prerequisite skill: Translate "sum," "product," and "less than" into symbols

Lesson 6-4: Write and Evaluate Numerical Expressions with Exponents

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.2.c

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

$4 + 6 \div 2$? $3 + 2^2 \times 5$? Which operation comes first?

Answer: $4 + 6 \div 2$: Divide first ($6 \div 2 = 3$), then add: $4 + 3 = 7$. $3 + 2^2 \times 5$: Exponent first ($2^2 = 4$), then multiply ($4 \times 5 = 20$), then add: $3 + 20 = 23$. Exponent comes first, then multiplication, then addition.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

$8 + 12 \div 4$? $5 + 3^2 \times 2$? Which operation comes first?

Answer: $8 + 12 \div 4$: divide first (3), then add: 11. $5 + 3^2 \times 2$: exponent (9), then multiply (18), then add: 23. The exponent comes first.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

$20 - 6 \times 2$? $4 + 2^3 \div 4$? Which operation comes first?

Answer: $20 - 6 \times 2$: multiply first (12), then subtract: 8. $4 + 2^3 \div 4$: exponent (8), then divide (2), then add: 6. The exponent comes first.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Evaluate $2 \times (3 + 4)^2 - 10$.

Answer: Parentheses: 7. Exponent: 49. Multiply: 98. Subtract: 88.

Check the result against the situation and explain your reasoning.

Lesson 6-5: Write and Evaluate Algebraic Expressions

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.2

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Evaluate $n + 8$ when $n = 6$.

Answer: 14

Replace n with 6: $6 + 8 = 14$.

Prerequisite skill: Substitute a number for a letter, then compute

TASK 2 • FOUNDATION 2

Evaluate $2x + 3$ when $x = 5$.

Answer: 13

Substitute, then multiply before adding: $2 \times 5 + 3$.

Prerequisite skill: Follow order of operations after substituting

TASK 3 • FOUNDATION 3

Evaluate $4n$ when $n = 7$.

Answer: 28

A number next to a variable means multiplication: 4×7 .

Prerequisite skill: Read $4n$ as $4 \times n$

TASK 4 • FOUNDATION 4

Write an expression for 9 more than a number n .

Answer: $n + 9$

More than indicates adding 9 to the starting number.

Prerequisite skill: Translate a word phrase into an expression

Lesson 6-5: Write and Evaluate Algebraic Expressions

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.2

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Evaluate $4n + 3$ when $n = 5$. Write an expression for "7 more than a number."

Answer: When $n = 5$: $4(5) + 3 = 20 + 3 = 23$. Expression for '7 more than a number': $n + 7$ (or $x + 7$).

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Evaluate $3n + 8$ when $n = 6$. Write an expression for "5 less than a number."

Answer: $3(6) + 8 = 18 + 8 = 26$. Expression: $n - 5$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Evaluate $2n - 7$ when $n = 10$. Write an expression for "the product of 6 and a number."

Answer: $2(10) - 7 = 20 - 7 = 13$. Expression: $6n$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Write an expression for the cost of n tickets at \$12 each plus a \$5 fee, then find the cost for 4 tickets.

Answer: $12n + 5$. At $n = 4$: $12(4) + 5 = 53$ dollars.

Check the result against the situation and explain your reasoning.

Lesson 6-6: Identify Equivalent Algebraic Expressions

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Simplify $5x + 2x$.

Answer: $7x$

Both terms count x units; combine coefficients $5 + 2$.

Prerequisite skill: Combine like terms such as $5x + 2x$

TASK 2 • FOUNDATION 2

Evaluate $3x + 4$ when $x = 2$.

Answer: 10

Substitute 2, then compute $6 + 4$.

Prerequisite skill: Evaluate an expression at a chosen value

TASK 3 • FOUNDATION 3

Evaluate $3(4 + 5)$ using distribution.

Answer: 27

$3 \times 4 + 3 \times 5 = 12 + 15$.

Prerequisite skill: Apply the distributive property with numbers, e.g. $3(4 + 5)$

TASK 4 • FOUNDATION 4

Compare $2(x + 3)$ and $2x + 6$ at $x = 4$. Give both values.

Answer: 14 and 14

$2(4 + 3) = 14$ and $2 \times 4 + 6 = 14$. Matching at one value alone does not prove equivalence for all values.

Prerequisite skill: Compare two results to decide whether expressions match

Lesson 6-6: Identify Equivalent Algebraic Expressions

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

$5x + 2x = ?$ Evaluate $2(x + 3)$ and $2x + 6$ at $x = 4$
— are they the same?

Answer: $5x + 2x = 7x$. At $x = 4$: $2(4 + 3) = 2(7) = 14$; and $2(4) + 6 = 8 + 6 = 14$. Yes, both equal 14 at $x = 4$. The distributive property, $2(x + 3) = 2x + 6$, shows they are equivalent for every x .

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

$8y + 3y = ?$ Evaluate $3(x + 2)$ and $3x + 6$ at $x = 5$
— are they the same?

Answer: $8y + 3y = 11y$. At $x = 5$: $3(7) = 21$ and $15 + 6 = 21$. Yes — the distributive property shows they match for every x .

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

$9a - 4a = ?$ Evaluate $4(x + 1)$ and $4x + 4$ at $x = 3$
— are they the same?

Answer: $9a - 4a = 5a$. At $x = 3$: $4(4) = 16$ and $12 + 4 = 16$. Yes — equivalent for every x .

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Are $3(2x + 4)$ and $6x + 7$ equivalent? Justify.

Answer: No. Distributing gives $6x + 12$, not $6x + 7$. At $x = 0$ they give 12 and 7 — different.

Check the result against the situation and explain your reasoning.

Lesson 6-7: Find Factors and Multiples

Connect & apply • 8 tasks • Prepares for Standard 6.NS.B.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute $84 \div 7$.

Answer: 12

The related fact is $7 \times 12 = 84$.

Prerequisite skill: Recall multiplication and division facts

TASK 2 • FOUNDATION 2

List every positive factor of 18.

Answer: 1, 2, 3, 6, 9, 18

Use factor pairs 1×18 , 2×9 , and 3×6 .

Prerequisite skill: List every factor pair of a number up to 100

TASK 3 • FOUNDATION 3

List the first four positive multiples of 7.

Answer: 7, 14, 21, 28

Multiply 7 by 1, 2, 3, and 4.

Prerequisite skill: List the first several multiples of a number

TASK 4 • FOUNDATION 4

Which of 2, 3, 5, 9, and 10 divide 135 evenly?

Answer: 3, 5, 9

Digit sum 9 makes it divisible by 3 and 9; final digit 5 makes it divisible by 5. It is not even and does not end in 0.

Prerequisite skill: Test divisibility by 2, 3, 5, 9, and 10

Lesson 6-7: Find Factors and Multiples

Connect & apply • 8 tasks • Prepares for Standard 6.NS.B.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

All the factors of 18? First five multiples of 6?
GCF of 18 and 24?

Answer: Factors of 18: 1, 2, 3, 6, 9, 18. First five multiples of 6: 6, 12, 18, 24, 30. GCF of 18 and 24: Common factors are 1, 2, 3, 6; Greatest is 6.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

All the factors of 24? First five multiples of 8?
GCF of 24 and 36?

Answer: Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24. Multiples of 8: 8, 16, 24, 32, 40. $\text{GCF}(24, 36) = 12$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

All the factors of 20? First five multiples of 5?
GCF of 20 and 30?

Answer: Factors of 20: 1, 2, 4, 5, 10, 20. Multiples of 5: 5, 10, 15, 20, 25. $\text{GCF}(20, 30) = 10$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

You have 24 pencils and 36 erasers to split into identical kits with nothing left over. What is the largest number of kits?

Answer: $\text{GCF}(24, 36) = 12$ kits, each with 2 pencils and 3 erasers.

Check the result against the situation and explain your reasoning.

Lesson 6-8: Generate Equivalent Expressions

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.3

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Find the greatest common factor of 18 and 30.

Answer: 6

Their common factors are 1, 2, 3, and 6; choose the greatest.

Prerequisite skill: Find the GCF of two numbers

TASK 2 • FOUNDATION 2

Factor $12x + 20$ using the greatest common factor.

Answer: $4(3x + 5)$

Both terms share 4; divide each term by 4 inside the parentheses.

Prerequisite skill: Apply the distributive property forward and backward

TASK 3 • FOUNDATION 3

Simplify $4a + 3a + 2$.

Answer: $7a + 2$

Combine the a terms; the constant 2 is a different kind of term.

Prerequisite skill: Combine like terms

TASK 4 • FOUNDATION 4

Compute $6(10 + 4)$.

Answer: 84

Distribute: $60 + 24 = 84$.

Prerequisite skill: Multiply a number by a sum

Lesson 6-8: Generate Equivalent Expressions

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.3

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

GCF of 18 and 24? Expand $3(x + 5)$. Factor $4x + 12$.

Answer: $\text{GCF}(18, 24) = 6$. Expand: $3(x + 5) = 3x + 15$. Factor: $4x + 12 = 4(x + 3)$.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

GCF of 20 and 30? Expand $5(x + 4)$. Factor $6x + 18$.

Answer: $\text{GCF}(20, 30) = 10$. $5(x + 4) = 5x + 20$. $6x + 18 = 6(x + 3)$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

GCF of 12 and 16? Expand $4(x + 7)$. Factor $9x + 15$.

Answer: $\text{GCF}(12, 16) = 4$. $4(x + 7) = 4x + 28$. $9x + 15 = 3(3x + 5)$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Factor $24x + 36$ completely, then check by expanding.

Answer: $\text{GCF}(24, 36) = 12$, so $24x + 36 = 12(2x + 3)$. Check: $12(2x) + 12(3) = 24x + 36$.
✓

Check the result against the situation and explain your reasoning.